

FINNER, Ireland

WMO: 039780

Lat: 54.4939N

Lon: 8.2431W

Elev: 114

StdP: 14.64

Time Zone: 1.00 (IRL)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	28.2	31.1	23.8	18.1	31.9	26.5	20.5	34.7	39.7	42.9	35.6	44.9	5.3	110	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	71.4	63.0	68.2	61.3	65.8	60.1	64.4	69.1	62.6	66.5	61.2	64.6	9.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.5	84.8	66.7	60.9	80.3	64.7	59.5	76.2	63.1	29.6	69.7	28.3	66.7	27.3	64.9	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.4	28.1	24.7	DB	25.6	77.0	3.9	3.5	22.8	79.5	20.5	81.6	18.3	83.6	15.5	86.2	(4)
(5)			WB	24.5	68.4	3.6	2.2	21.9	70.0	19.8	71.3	17.8	72.5	15.1	74.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	42.8	42.4	44.4	47.7	52.1	56.7	58.9	58.6	56.1	51.4	46.3	43.2	(6)	
	DBStd	7.56	5.05	4.81	4.81	4.36	4.61	4.12	3.52	2.67	3.64	4.50	4.87	5.77	(7)	
	HDD50	1131	226	215	184	94	26	1	0	0	2	37	128	218	(8)	
	HDD65	5456	688	632	640	520	401	253	194	200	268	421	562	677	(9)	
	CDD50	1159	3	3	9	24	91	201	275	265	185	81	16	6	(10)	
	CDD65	7	0	0	0	0	0	2	4	1	0	0	0	0	(11)	
	CDH74	46	0	0	0	0	5	19	21	1	0	0	0	0	(12)	
	CDH80	4	0	0	0	0	0	2	2	0	0	0	0	0	(13)	
	Wind	WSAvg	11.8	13.8	12.7	12.7	11.4	11.0	10.3	10.0	10.8	11.5	11.5	12.5	13.0	(14)
	Precipitation	PrecAvg	55.0	5.9	4.6	4.0	3.6	3.4	4.1	4.3	4.1	5.5	5.9	6.3	(15)	
(16)		PrecMax	64.3	8.5	9.8	7.6	6.1	7.4	6.9	8.3	6.6	10.5	11.7	12.0	(16)	
(17)		PrecMin	41.3	0.7	1.5	1.6	1.4	0.8	1.1	2.0	0.7	0.9	3.2	2.0	(17)	
(18)		PrecStd	6.5	2.0	2.2	1.7	1.3	1.7	1.4	1.4	1.7	1.5	2.5	2.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	55.9	60.3	65.2	73.9	76.0	76.7	70.9	69.3	64.0	59.4	57.1	(19)	
		MCWB	52.7	51.7	52.1	54.4	62.6	63.5	66.5	64.0	62.2	59.4	56.1	53.9	(20)	
	2%	DB	53.2	53.5	56.0	60.3	67.8	70.5	70.9	67.2	66.1	61.1	56.6	55.0	(21)	
		MCWB	51.0	50.4	50.5	52.5	59.3	61.0	64.1	61.8	60.7	57.5	54.1	52.0	(22)	
	5%	DB	51.8	51.7	53.6	57.2	63.3	66.9	67.5	65.1	63.9	59.4	54.7	53.4	(23)	
		MCWB	49.5	49.1	49.5	51.0	56.1	59.8	62.0	60.6	59.8	56.1	52.5	50.9	(24)	
	10%	DB	50.2	49.8	51.6	54.6	60.0	63.7	64.8	63.4	61.9	57.8	53.1	51.4	(25)	
		MCWB	48.0	47.2	48.0	49.7	54.2	58.2	60.0	59.4	58.3	54.8	50.8	49.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	52.6	53.3	56.2	63.7	65.6	68.4	65.5	64.0	60.3	56.8	54.4	(27)	
		MCDB	55.1	54.7	57.8	61.9	71.5	73.8	74.9	68.9	66.4	63.0	58.9	56.5	(28)	
	2%	WB	51.5	50.9	51.5	53.8	60.2	62.6	64.9	63.2	61.8	58.3	54.5	52.5	(29)	
		MCDB	53.1	53.0	54.7	58.6	66.9	68.9	69.7	65.9	64.6	60.5	56.0	54.5	(30)	
	5%	WB	49.8	49.3	50.0	51.8	57.4	60.6	62.7	61.7	60.3	56.7	52.8	51.1	(31)	
		MCDB	51.4	51.5	53.1	56.4	61.9	65.6	66.5	64.3	63.0	58.9	54.4	53.1	(32)	
	10%	WB	48.3	47.4	48.4	50.1	54.9	58.8	60.8	60.2	58.9	55.2	51.3	49.3	(33)	
		MCDB	49.9	49.6	51.2	53.9	58.8	62.9	64.0	62.7	61.5	57.4	53.0	51.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	8.8	10.4	11.5	11.9	11.1	9.9	9.2	9.7	9.3	8.2	8.1	(35)	
		MCDBR	8.9	8.9	12.7	15.4	17.7	17.1	14.3	11.5	12.0	9.9	8.6	9.6	(36)	
		MCWBR	8.8	7.9	8.8	9.3	10.5	10.1	9.0	7.6	8.4	7.6	8.3	8.8	(37)	
	5% WB	MCDBR	8.5	8.8	11.2	13.3	15.8	15.1	13.4	10.1	10.7	8.9	8.3	9.0	(38)	
		MCWBR	8.7	8.3	8.8	9.0	10.3	9.5	9.0	8.0	8.4	7.9	8.4	8.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.347	0.353	0.374	0.383	0.391	0.377	0.380	0.364	0.362	0.349	0.338	0.336	(40)	
	taud		2.197	2.277	2.271	2.290	2.311	2.388	2.389	2.446	2.443	2.398	2.263	2.170	(41)	
	Ebn at Noon		178	228	251	267	271	276	273	271	255	229	182	153	(42)	
	Edh at Noon		21	26	33	36	38	36	35	31	28	23	19	17	(43)	
(44) All-Sky Solar Radiation	RadAvg		179	397	739	1199	1533	1565	1400	1172	829	475	221	130	(44)	
	RadStd		18	36	63	99	130	139	96	94	49	38	20	13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page